

Cross-Domain Misinformation Detection using Affective Information and In-Context Learning – a RAEmoLLM extension

Antonis Prodromou, Orestis Georgiadis
MSc Data and Web Science, Aristotle University of Thessaloniki

INTRODUCTION

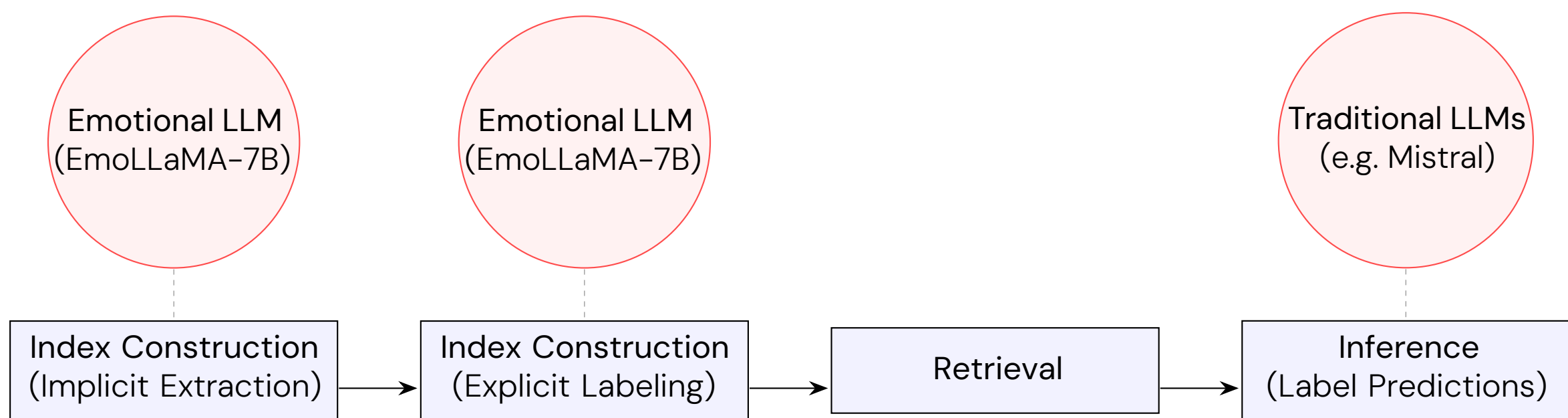
Social media enables the rapid spread of misinformation, damaging public trust and causing real-world harm. Manual fact-checking cannot keep pace with the volume of content.

In this project, we validate through experiments RAEmoLLM, a Retrieval-Augmented Generation (RAG) framework that uses emotional cues in text to detect misinformation across different topics using few-shot prompts in transformer Natural Language Processing (NLP) models.

We then extend the framework to also account for the certainty degree expressed in texts assigning them a certainty score ('Cert'), which takes continuous values between [0,1].

METHODOLOGY

Framework Architecture:



1. Index Construction

• An Emotional LLM calculates emotional intensity scores for tweets and generates affective embeddings (4096-dimensions) for each, storing them in a database.

2. Retrieval Module

- Computes cosine similarity between the target tweet and the database, using implicit information only.
- Selects the top-4 most emotionally similar examples and uses these as few-shot demonstrations for the LLM.

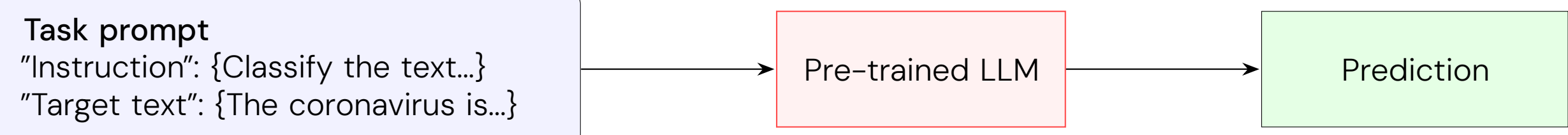
3. Inference Module

• Applies a traditional transformer to classify tweets as Unrelated, Related, or Conspiracy, using explicit labelling on the target and the few-shot examples, i.e. the scores calculated at Index Construction.

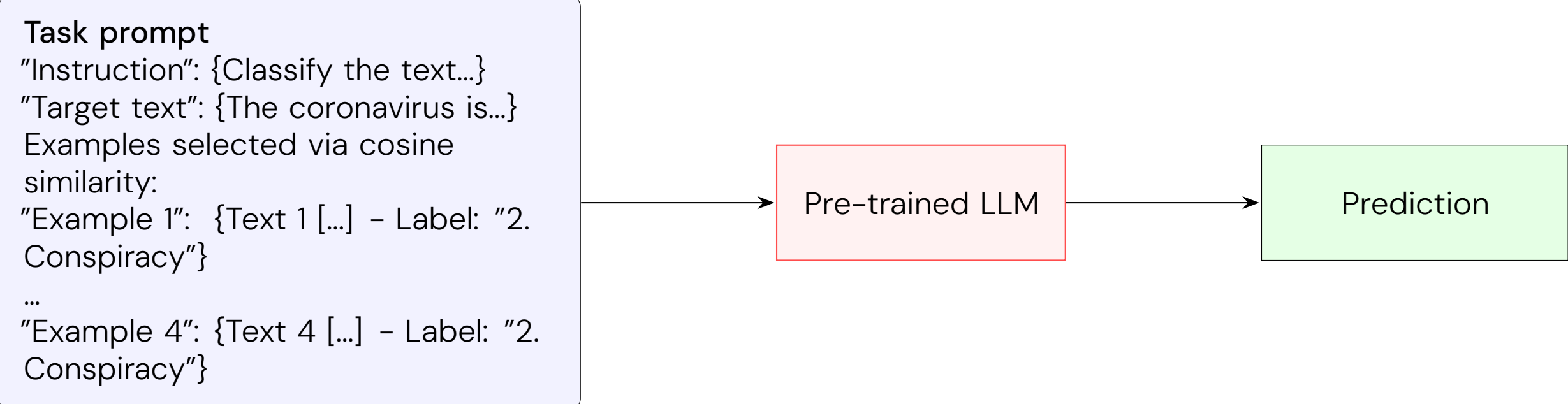
EXPERIMENTAL MODELS

- Zero-shot baseline (no examples).
- RAG with Explicit Sentiment Valence Intensity (Vreg) only.
- RAG with Explicit Vreg + Certainty degree (concatenation).
- RAG with Explicit Vreg + Certainty degree (score averaging).

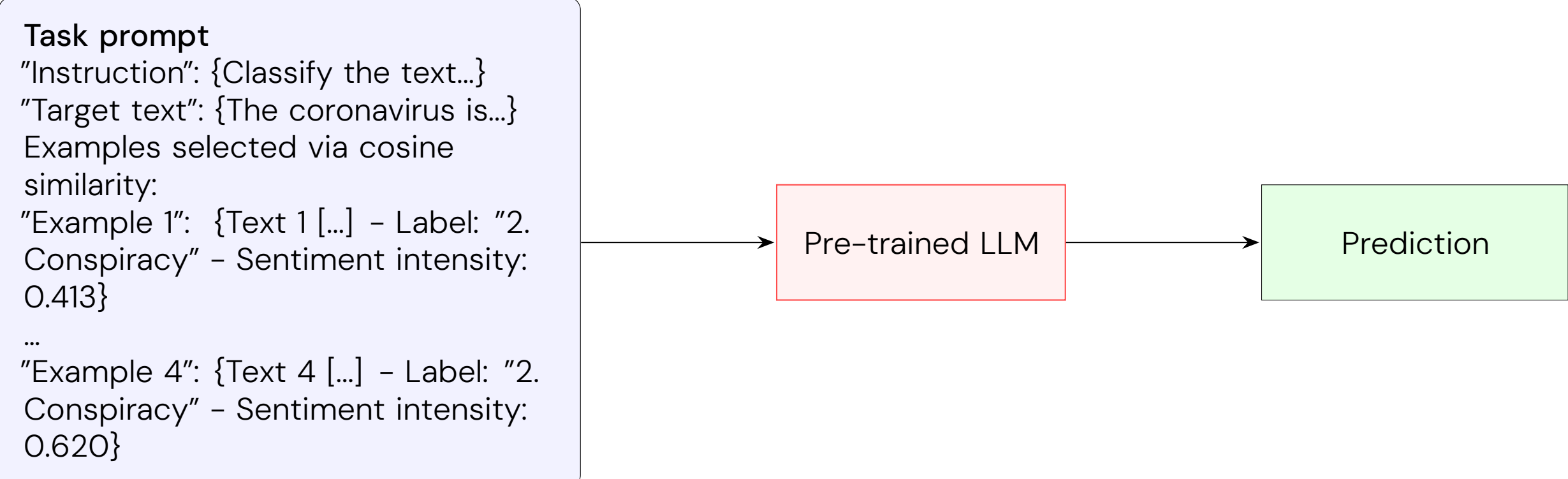
Zero-Shot Inference



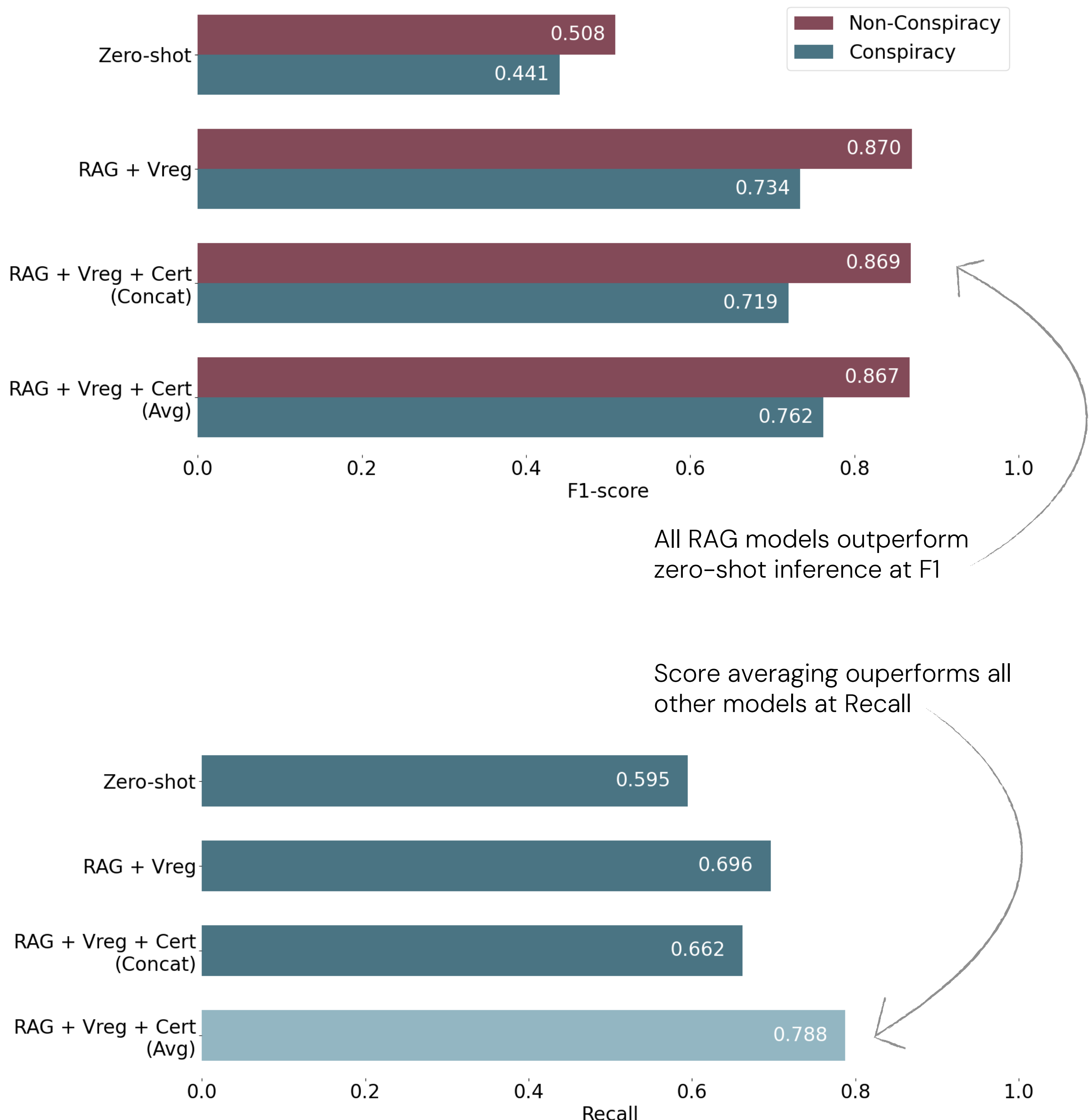
Few-Shot, Implicit Inference



Few-Shot, Explicit Inference



RESULTS



WHY SCORE AVERAGING WORKS

- Maintains separate 4096-dimensional embeddings for each affective dimension (Valence intensity and Certainty degree).
- Combines similarity scores linearly: $\text{Score} = 0.5 \times \cos(\text{Vreg}) + 0.5 \times \cos(\text{Cert})$.
- Prevents information loss from averaging embeddings.

CONCLUSION

We successfully reproduced and extended the RAEmoLLM framework, demonstrating that affective information significantly improves misinformation detection. The project confirms that:

- Affective cues serve as a "latent space" for detecting misinformation without the need for model fine-tuning.
- Certainty can be used as an affective dimension for misinformation detection.
- Score averaging for few-shot retrieval outperforms embedding concatenation.

FUTURE WORK

Testing the framework on multi-modal content (images + text) and expanding the affective dimensions beyond valence intensity and certainty.

REFERENCES

Key source: Liu et al. (2024) RAEmoLLM: Retrieval Augmented LLMs for Cross-Domain Misinformation Detection Using In-Context Learning Based on Emotional Information.

ACKNOWLEDGMENTS

This research was conducted using the COCO dataset provided by Langguth et al. (2022) (3,495 tweets across 12 conspiracy categories).

